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Secure Contract-Farming Platform with MERN Stack and Machine Learning

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OUTLINE

INTRODUCTION

- Agriculture is one of the largest employment sectors in India, but farmers still face major challenges like **lack of transparency and trust** in contract farming.

- Farmers often receive **unfair prices** due to manual grading and middlemen involvement.

- Traditional systems do not provide:
    - Secure contracts
    - Proper verification
    - Transparent quality evaluation

To address these challenges, this paper proposes **FarmLink**, a secure and scalable web-based platform built using the **MERN stack**. The system integrates **AI-based crop quality classification** to provide objective and data-driven evaluation of agricultural produce.

INTRODUCTION

- The system is built using:
  MERN Stack (MongoDB, Express, React, Node.js)
  AI-based crop quality prediction

- FarmLink improves:
  Transparency in contracts
  Trust between farmer and buyer
  Fair pricing using AI

- The platform also ensures **security using JWT authentication and encryption techniques**, making contract data safe and reliable.

OBJECTIVE

- This paper has multiple key objectives. First, it proposes the development of a **secure and transparent digital platform for contract farming** using the MERN stack. The system enables structured interactions between farmers and buyers through **contract creation, negotiation, and agreement execution**.

- Second, the paper focuses on building a **complete contract lifecycle management system** that includes **tracking, verification, and dispute resolution**. This ensures accountability at every stage, from cultivation to final delivery, thereby improving trust and reducing conflicts.

- Additionally, the platform integrates important features such as **voice-enabled harvest listing, multi-language support, real-time notifications, and secure communication**. These features enhance usability and make the system suitable for real-world agricultural scenarios.

PROBLEM STATEMENT

- The main concern in contract farming is the lack of transparency and trust between farmers and buyers.
- Unfair pricing and subjective crop evaluation often lead to financial loss for farmers.
- Lack of proper contract enforcement and verification can result in disputes and misuse of agreements.
- In existing systems, there is no structured way to track contract progress from cultivation to delivery.
- With increasing digital adoption, ensuring secure communication and reliable data handling becomes essential.
- An efficient system is required to provide transparency, accountability, and fair decision-making in contract farming.

LITERATURE REVIEW

- **S. Kumar [1]** discusses contract farming in India and highlights both its opportunities and challenges. The study shows that although contract farming can improve farmer income, lack of proper regulation and transparency leads to unfair practices.

- **R. Singh [2]** explores the use of mobile-enabled technologies in agriculture. The study emphasizes that digital platforms can support farmers with better information access and decision-making, reducing reliance on intermediaries.

- **A. Sharma and P. Mishra [3]** focus on machine learning-based prediction systems in agriculture. Their work demonstrates how data-driven approaches can improve accuracy, but real-world implementation challenges still exist.

LITERATURE REVIEW

- **G. Bhosale [4]** presents an IoT-based crop monitoring system to improve agricultural productivity. The study highlights the importance of real-time monitoring while noting limitations related to infrastructure and scalability.

- **M. Koklu and I. Ozkan [8]** introduce a rice image dataset for crop classification using computer vision techniques. Their work supports AI-based quality assessment, helping replace subjective grading methods.

- **G. Ton et al. [7]** analyze the impact of contract farming on farmer income. The study concludes that transparency and proper system design are essential for achieving fair outcomes.

SYSTEM MODEL

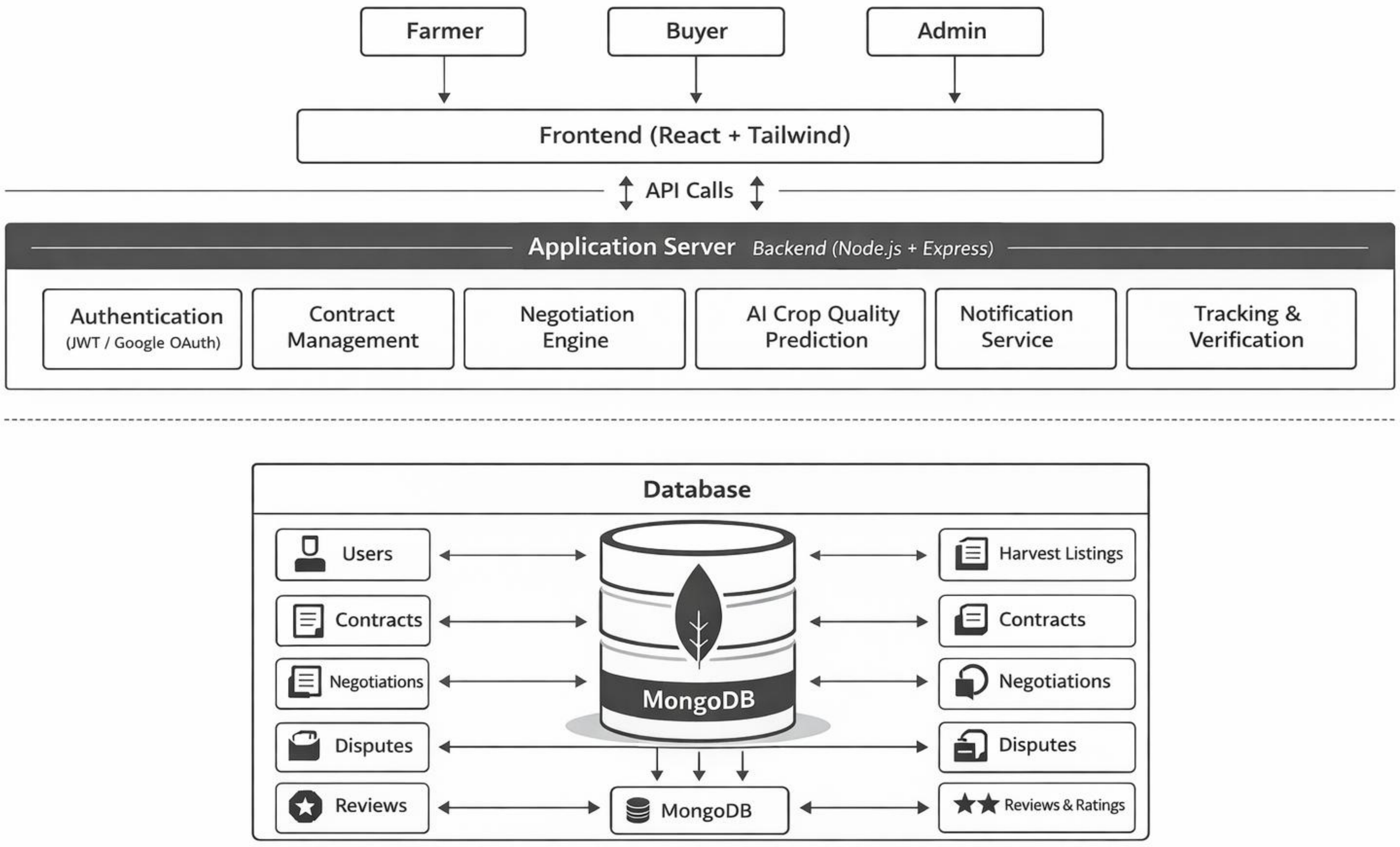

Fig 1. System Model of Farmlink

SYSTEM MODEL

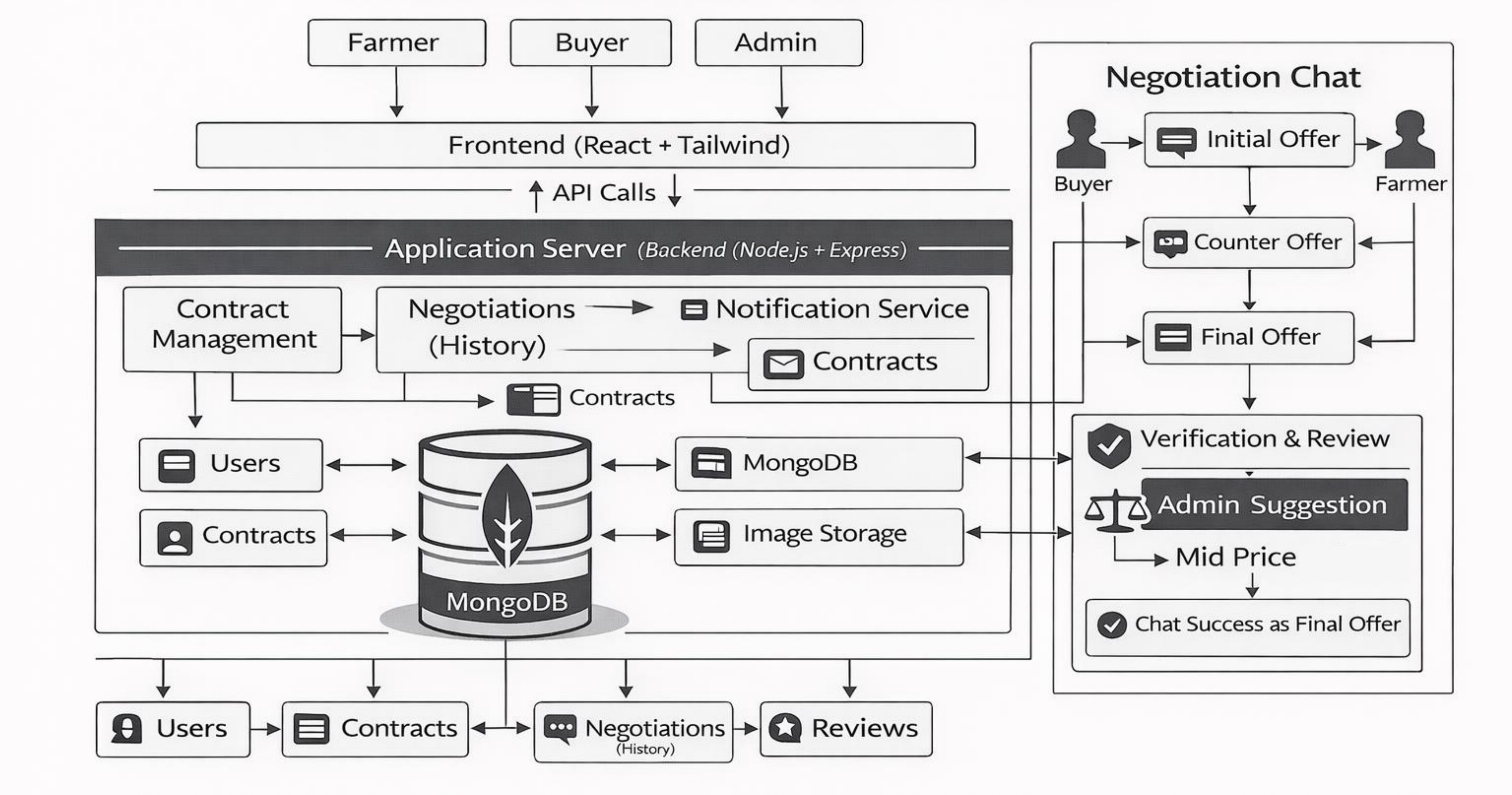

Fig 2. Contract Management

SYSTEM ARCHITECTURE

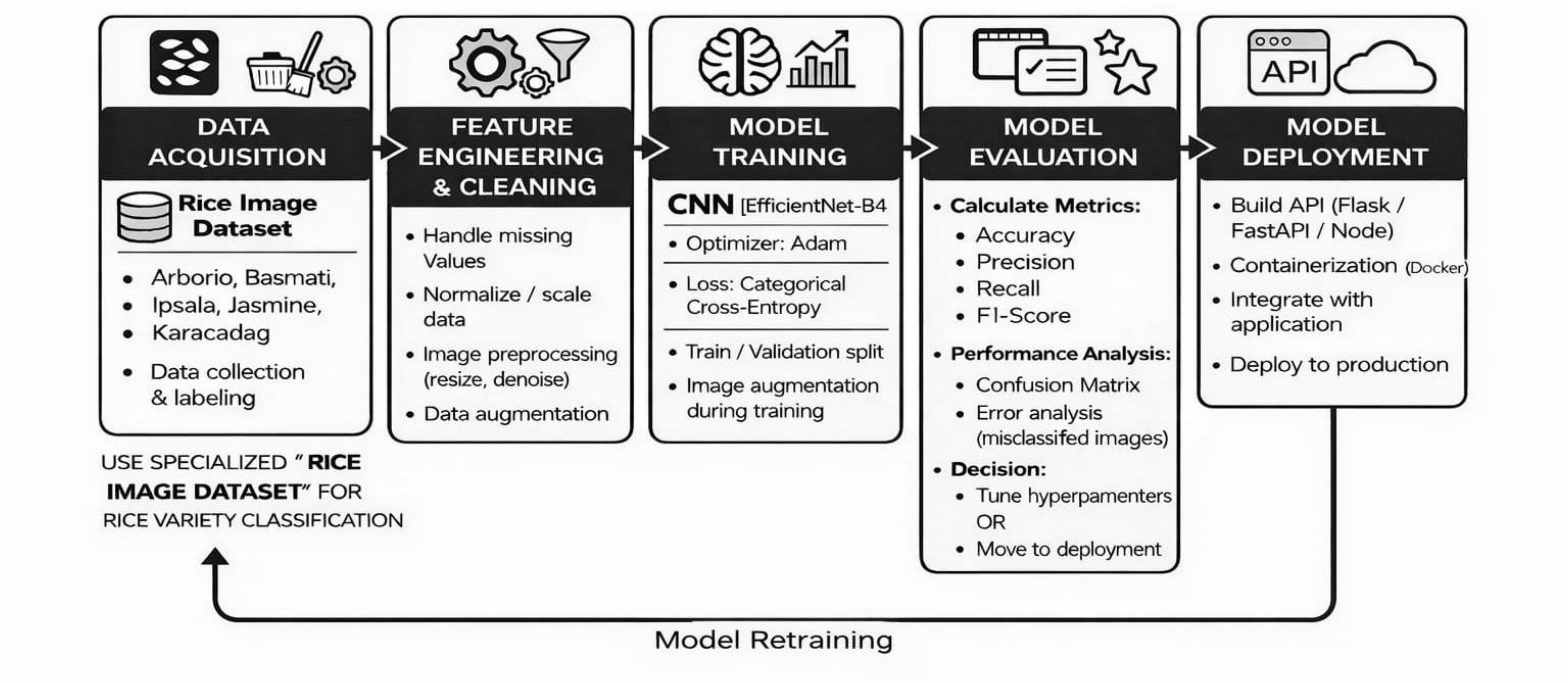

Fig 3. AI Model System Architecture

IMPLEMENTION DETAILS

- The proposed system is developed using the **MERN stack**, where React is used for frontend, Node.js and Express for backend, and MongoDB for data storage.
- The platform provides role-based dashboards for **farmers, buyers, and admins**, enabling contract creation, negotiation, and management.
- Secure authentication is implemented using **JWT and Google OAuth**, ensuring protected access to the system.
- The contract lifecycle includes **creation, negotiation, acceptance, tracking, and verification**, maintaining transparency at each stage.

IMPLEMENTION DETAILS

- Real-time features such as **notifications, chat, and live tracking** are integrated to improve communication and monitoring.
- The system also includes **AI-based crop quality prediction**, supporting objective evaluation of agricultural produce.
- Additional features like **voice-enabled input and multi-language support** enhance usability for rural users.

RESULT EVALUATION

The proposed FarmLink system was implemented as a secure digital platform for contract farming using the MERN stack. The evaluation is based on system functionality, model performance, and insights derived from referenced studies.

Aspect	Traditional System	FarmLink System
Quality Evaluation	Subjective	AI-based
Contract Transparency	Low	High
Tracking	Not Available	Stage-wise Tracking
Dispute Handling	Weak	Structured
Accessibility	Limited	Multi-language & Voice

Table 1: System Advantages Over Traditional Methods

RESULT EVALUATION

- classification module demonstrated a high learning stability on the Murat Koklu Rice Image Dataset. The training accuracy was 96.38, and the validation accuracy was 99.11, The high final training loss (0.1102) was a good indication of the reliability of the model in real world produce grading.

Metric	Definition	Measurement Method	Target Value
Classification Accuracy	Precision of crop variety and quality grading.	Ratio of correct predictions over total images in the validation set.	**>=96%**
Trust Index	Composite score reflecting contract transparency and satisfaction.	Likert-scale survey responses normalized to a 0–1 range.	**>=0.80%**
Income Change (%)	Percentage increase in average farmer income after platform adoption.	Comparison of baseline income vs. post-adoption outcomes.	**>= 12%**
System Latency	**Average response time of the ML service per request.**	**API response time measured under normal operating conditions.**	**< 150 ms**

Table 2:Evaluation Metrics and KPI Targets

CONCLUSION

- The proposed FarmLink system presents a **secure and scalable solution** to address challenges in contract farming. By integrating **structured contract management with digital technologies**, the platform improves **transparency, accountability, and trust** between farmers and buyers.

- The MERN-based architecture enables efficient system performance, while features like **negotiation, tracking, and dispute resolution** ensure a complete and reliable contract lifecycle.

- Additionally, **AI-based crop quality prediction** supports objective evaluation, reducing manual dependency and improving pricing fairness. Features such as **multi-language support and voice-enabled input** enhance usability for real-world adoption.

- Overall, the system demonstrates strong potential to build a **transparent and reliable digital ecosystem**, reducing disputes and improving outcomes for farmers.

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THANK YOU...